RECEIVER (RX) = FFE → DFE → SLICER → GRAY LABELS + ADAPTATION

P16 x[n]
3 × Q12 / R

P15 ψ[n] / g[n]

TWO COEFFICIENT UPDATE BRANCHES — SAME CURRENT ERROR AND GATE

P12 aligned TX reference → training input

FFE LMS UPDATE

Inputs: P16 x, P15 ψ/g, P18 A_w
Δ_w=round_F20(μ_Q ψ x)
A_w_next=P_W(A_w + Δ_w)
g=0 → hold current state
P_W projects each tap to W
OUT P17 A_w_next[n]
3 × signed40 F20 / W

P17 / F20 / W

z⁻¹ / FFE COEFFICIENT STATE

P18 A_w[n]=A_w_next[n−1] / F20 / W
P19 w[n]=Q12(A_w[n]); 3 × Q12 / W
Initial state: [0,0.75,0]

P19 w[n] / W

DFE LMS UPDATE

Inputs: P11 d_prev, P15 ψ/g, P21 A_b
Δ_b=round_F20(μ_Q ψ d_prev)
A_b_next=P_B(A_b − Δ_b)
g=0 → hold current state
P_B projects the tap to B
OUT P20 A_b_next[n]
signed40 F20 / B

P20 / F20 / B

z⁻¹ / DFE COEFFICIENT STATE

P21 A_b[n]=A_b_next[n−1] / F20 / B
P22 b0[n]=Q12(A_b[n]); Q12 / B
Initial state: 0

P22 b0[n] / B

P15 ψ / g

P11 / Q12 / S0

TRAINING / DD SELECTOR

P12 d_ref during n≤3000
P09 d_hat after n>3000
OUT P13 d_des[n]: Q12 / S0
Startup n≤D: d_des=0

P13

C1 / LMS ERROR COMPARISON

e_raw[n]=satQ(d_des[n]−y[n])
Inputs: P13 desired; P08 y (pre-slicer)
OUT P14 e_raw: Q12 / Q
This comparison drives adaptation.

P14

P09 d_hat[n] / S4

SHARED ROBUST ERROR / UPDATE CONTROL

ψ[n]=clip(e_raw[n],−2.5,+2.5); OUT P15 ψ: Q12 / [−2.5,+2.5]
g[n]=1 in training; in DD g=(|e_raw[n]|≤1.5); OUT P15 g∈{0,1}
Both LMS branches use this same ψ[n] and g[n].
No extra error-path z⁻¹ is inserted.

P08 y[n] — tap BEFORE the slicer

P06 r[n]
Q12 / R

P07 f[n] / wide F12
Bnd: [−22.499268,+22.498535]

FFE / 3 TAPS

Inputs: P06 r[n], P19 w[n]
x[n]=(r[n],r[n−1],r[n−2])
Two internal z⁻¹ sample memories
f[n]=Σ qmul(wk,xk), k=0..2
OUT P07 f[n]: wide signed F12
Bnd: [-22.499268, 22.498535]
Obs: [-5.3218, 5.3542]
P16 x[n] to FFE LMS: 3 × Q12 / R

DFE / 1 TAP + SUBTRACT

Inputs: P07 f, P11 d_prev, P22 b0
v[n]=qmul(b0[n],d_prev[n])
Feedback v Bnd: [−2.25,+2.25]
y[n]=satQ(f[n]−v[n])
OUT P08 y[n]: Q12 / Q
Obs: [-3.9556, 3.9580]

P08 y[n]
Q12 / Q

PAM-4 SLICER

Input P08: pre-slicer y[n]
Thresholds: −2, 0, +2
Exact threshold → lower level
OUT P09 d_hat[n]: Q12 / S4
Examples: y=0 → −1; y=2 → +1

P09 d_hat[n]
Q12 / S4

GRAY LABELS

Symbol → Gray bits
−3/−1/+1/+3 → 00/01/11/10
OUT P10: (d_hat[n], b_hat[n])
S4 × {00,01,11,10}
Measurement postprocessing

P09 / S4

z⁻¹ / DECISION MEMORY

OUT P11: d_prev[n]=d_hat[n−1]
Q12 / S0; initialized to zero

P11 d_prev[n]
Q12 / S0

DETAIL RANGE LEGEND

W=[−1.25,+1.25]; B=[−0.75,+0.75]
Q12: signed16/F12; Q=[−8,+7.999755859375]
R=[−5.999755859375,+5.99951171875]
S4={−3,−1,+1,+3}; S0=S4 ∪ {0 at startup}
F20: signed40/F20; ψ∈[−2.5,+2.5], g∈{0,1}